

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Zoology**Paper: ZOB-504 — Biotechniques**

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:(a) Select and write the best options in the following: [4 × 0.5 = 2]

(i) In gel electrophoresis, the detergent sodium dodecyl is used to:

- (1) adjust the pH of protein
- (2) renature protein
- (3) make the protein positively charged
- (4) make the protein negatively charged

(ii) Which is the stain commonly used in electron microscopy?

- (1) Uranyl acetate
- (2) Lead citrate
- (3) Both (a) and (b)
- (4) None of the above

(iii) The pH indicator in animal cell culture medium is:

- (1) methyl red
- (2) phenol red
- (3) methyl orange
- (4) All of the above

(iv) Formalin solution (10% unbuffered) is composed of:

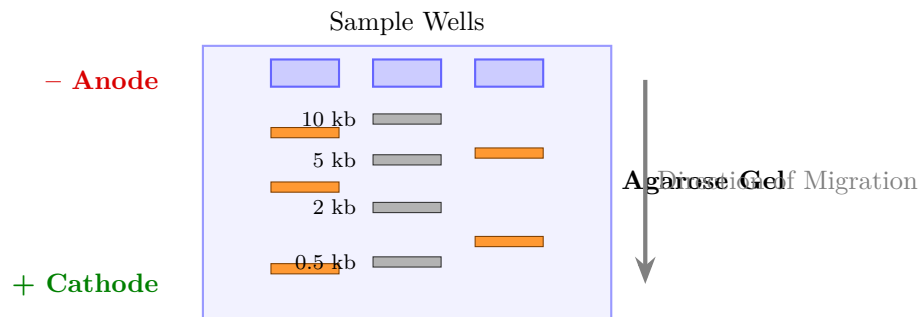
- (1) 10 ml formaldehyde (40%) and 90 ml distilled water
- (2) 50 ml formaldehyde (40%) and 50 ml distilled water
- (3) 10 ml formaldehyde (10%) and 90 ml distilled water
- (4) 50 ml formaldehyde (10%) and 50 ml distilled water

(b) State whether the following statements are True or False. Justify your answer in brief:
[3 × 2 = 6]

- (i) In a phase-contrast microscope, the undiffracted or background light is retarded by the phase plate.
 - (ii) The wavelength of electrons is dependent on the accelerating voltage.
 - (iii) Differential centrifugation is based on the difference in the size of biological particles of different densities.
- (c) Define the following: [4 × 1 = 4]
- (i) R_f value
 - (ii) Microtome
 - (iii) Critical point drying
 - (iv) Limit of resolution
- (d) Expand the following: [4 × 0.5 = 2]
- (i) EDTA
 - (ii) DMEM
 - (iii) TEMED
 - (iv) LASER
- (e) Differentiate between the following: [2 × 3 = 6]
- (i) Bright field and Dark field microscope
 - (ii) Paper and Thin layer chromatography

Question 2. Write down the principle of centrifugation. Discuss the applications of analytical centrifugation. [10]

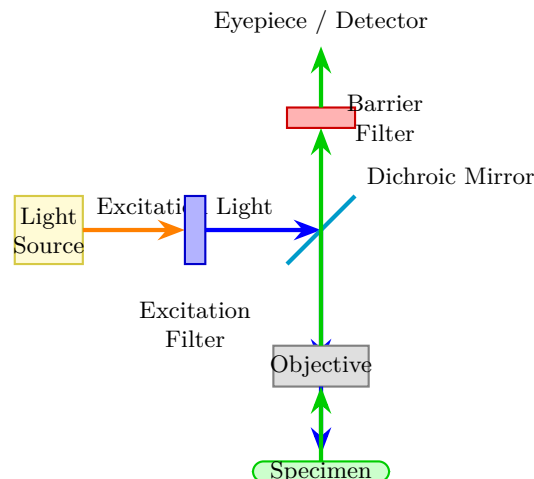
Question 3. What do you understand by electrophoretic mobility? Discuss the procedure of electrophoretic separation of nucleic acids. [10]



Question 4. Answer the following:

- Write down the factors that affect the fixation of biological tissue samples. [5]
- Describe briefly the hematoxylin and eosin staining methods of paraffin-embedded tissue sections. [5]

Question 5. Describe with a labeled diagram, the working principles and applications of the fluorescence microscope. [10]



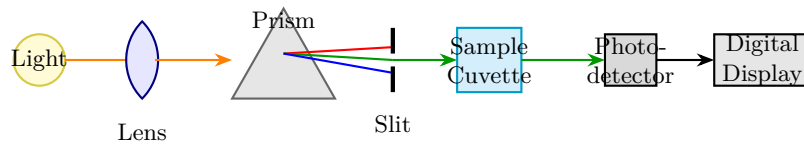
Question 6. Define radioactivity. Explain the various methods used for the detection and measurement of radioactivity. [10]

Question 7. Describe different types of animal cell culture. Write down a stepwise procedure to initiate culture from animal tissues. [10]

Question 8. Write short notes on the following: [2 × 5 = 10]

(a) Cell viability test

(b) Spectrophotometer



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